Name: Cohort/Batch: Date: Score:

NEXT.JS PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Web Technologies

TOTAL: 10 QUESTIONS

Q1 What is Next.js and what problem in Web Technologies does it solve? Fundamental

Q2 What are the core components or concepts in Next.js? Architecture

Q3 How does Next.js compare to its main alternatives? Configuration

Q4 How do you get started with Next.js for a new project? Hardening

Q5 What configuration options most affect Next.js's behavior in production? OAuth IdP

Q6 How does Next.js handle reliability and high availability? Observability

Q7 What observability does Next.js provide out of the box? Lifecycle

Q8 What are common performance bottlenecks in Next.js? Compliance

Q9 What security best practices apply when running Next.js? Testing

Q10 What mistakes do teams commonly make when adopting Next.js? Tuning

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 **Next.js is a tool or platform that** — Mitigation

Next.js is a tool or platform that automates or simplifies a specific concern in the Web Technologies domain, reducing manual effort and operational complexity for engineering teams.

A2 **Next.js has key building blocks that work** — Primitives

Next.js has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 **Next.js trades off simplicity against feature richness** — Hardening

Next.js trades off simplicity against feature richness differently than its alternatives. Choose Next.js when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 **Install Next.js via its package manager or** — Gotchas

Install Next.js via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 **Production-critical settings typically include concurrency** — Integration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 **Next.js provides HA through leader election, replication,** — Observability

Next.js provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 **Next.js exposes metrics (Prometheus endpoint or built-in** — Automation

Next.js exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 **Performance bottlenecks typically stem from misconfigured** — Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 **Run Next.js with the principle of least** — Assessment

Run Next.js with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 **Common pitfalls include skipping the documentation and** — Standardization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.